Design of a Fuzzy Decision-making Model and Its Application to Software Functional Size Measurement

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Abstract

COSMIC-FFP as a Functional Size Measurement method has greatly improved the estimation of real-time software system. Based on math expectation and variance, this paper puts forward the concepts of fuzzy expectation and fuzzy variance. And then the fuzzy regression analysis is formed on the basis of these concepts. The effort estimation decision-making model, which can determine regression frequency according to actual needs, is established based on this fuzzy regression analysis. This effort estimation decision-making model can also make the disturbance of morbid data and bad information in regression analysis to the decision result decrease bit by bit until we are satisfied. When applying this estimation model to software metrics of COSMIC-FFP, we can estimate the effort accurately, which leads to a correct decision.

1. Introduction

Software practitioners have realized the importance of realistic estimates of effort to the successful management of software projects. Estimates are necessary throughout the whole development life cycle. They are fundamental when people bid for a contact or determine a project's feasibility in terms of cost-benefit analysis. In addition, they allow project managers and development organizations to manage resources effectively.

Size, which can be described in terms of length, functionality and complexity, is often a major determinant factor of effort. Although length and complexity are also essential aspects of size, most effort prediction models concentrate on functional measurement of size.

Functional Size Measurement (FSM) methods are used to measure the size of software by quantifying the Functional User Requirements (FUR) of the software. A FSM method measures the logical external view of the software from the user's perspective by evaluating the amount of functionality to be delivered [1][2]. The capability to accurately quantify the size of software in an early or late stage of the development lifecycle is critical to software project managers. COSMIC-FFP (Common Software Measurement International Consortium Full Function Point) is a candidate method of FSM methods [4]. The recent development of COSMIC-FFP as a functional size measurement metric has greatly improved the functional sizing of real-time systems [3]. Furthermore, fuzzy logic models have been widely applied in estimation because their fuzzy output reduces the reliance on a single value for estimation, which can lead to loss of resources if the value is erroneous.

In spite of the presence of many fuzzy logic estimation models [5], none is adaptive for a COSMIC-FFP estimation model. The main contribution of this paper is to establish a fuzzy decision-making model for the COSMIC-FFP data [6]. In this paper, the concepts of fuzzy expectation and fuzzy variance are put forward based on math expectation and variance. These concepts are taken as theoretical bases to put forward fuzzy regression analysis and establish a practical model of fuzzy decision-making. It has been proved that this estimation model is practical.

This paper is structured as follows. Section 2 focuses on Fuzzy expectation and fuzzy variance. The fuzzy regression model based on the concepts of fuzzy expectation and fuzzy variance are established in Section 3. In section 4, the fuzzy decision-making model is applied to the COSMIC-FFP data set [6]. The preliminary results of this application using fuzzy regression analysis are presented in Section 5 and conclusions are drawn in Section 6.

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2. Fuzzy expectation and fuzzy variance

2.1. Fuzzy expectation

Math expectation describes an important characteristic of random variables and can be used to estimate the value of random variable. But in practice, because the data is not continuous and finite, it is impossible to gain exact probability values. So fuzzy expectation will be the realistic math model. Since the probability values are unknown, suppose that random variables are equal probability and then the math expectation can be expressed as [5][8]:

$$E(X) = \frac{1}{n}\sum_k x_k \qquad (1)$$

Here, n denotes the number of sample points. And then the math variance can be expressed as [5][8]:

$$D(X) = \frac{1}{n}\sum_k (x_k - E(X))^2 \qquad (2)$$

Here, $E(X)$ is the mean value of random variables (i.e. the math expectation of an equal probability event) and $D(X)$ is the variance of an equal probability event. So fuzzy expectation is defined as:

$$E_F(X) = \frac{\sum_k u_k x_k}{\sum_k u_k} \qquad (3)$$

Here, u_k is the membership function that the fuzzy set corresponds to [5]. The value of u_k ($u_k \in (0,1)$) corresponds to the values of random variables. But in practical applications, the value of u_k may be 0 or 1. If the value of u_k in expression (3) is 1, the expression (3) will become:

$$E_F(X) = \frac{1}{n}\sum_k x_k = E(X) \qquad (4)$$

From the above formula, we can see that when the membership degree is 1, fuzzy expectation will be the mean value of random variables and also be the math expectation of equal probability events.

2.2. Fuzzy variance

According to the definition of fuzzy expectation, fuzzy variance is defined as:

$$D_F(X) = \frac{\sum_k u_k (x_k - E_F(X))^2}{\sum_k u_k} \qquad (5)$$

In the same way, when $u_k = 1$, then $E_F(X) = E(X)$ and the expression (5) will become:

$$D_F(X) = \frac{1}{n}\sum_k (x_k - E_F(X))^2 = D(X) \qquad (6)$$

Here, fuzzy variance is obviously the variance of an equal probability event.

3. Fuzzy regression model

3.1. Fuzzy regression analysis

Fuzzy regression methods have considerably widened the field of application of classical regression methods in such a way that they are helpful in determining regression relations from fuzzy initial data, which can be either quantitative or qualitative. Therefore, fuzzy regression methods can be applied to expert estimation and several fields of human activities. When applying regression analysis to predict random variables based on the existing observed values, we can see that the data obtained will be affected more or less by random factors. Suppose that n sample points have been gained, so if all data points approximately reflect the actual things, the result gained will be relatively reasonable. Such a result is very realistic but seldom appears in practice. So when one or some data points are greatly influenced by random factors, there must be a difference between the result gained and the actual values. At that time, the classical regression analysis is not applicable.

In this paper, fuzzy least squares method is put forward based on fuzzy expectation and fuzzy variance. By fuzzy least squares method, the regression parameters are estimated and the corresponding regression model is established. Such is taken as fuzzy regression analysis. Fuzzy regression analysis can make the disturbance of morbid data and bad information to the decision result decrease to the minimum and raise accuracy and reliability of the decision result accordingly.

3.2. Establishing the fuzzy regression model

3.2.1. Determining dependent and independent variables

If a variable y is linearly relative to other variables $x_1, x_2, \ldots, x_k$, the regression model of these variables is represented as [7]

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k \qquad (7)$$

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Here, variables $b_0, b_1, b_2, \ldots, b_k$ are $k+1$ regression parameters to be determined later. When the $k+1$ parameters are determined based on the actually observed values, the regression model will be determined accordingly and y will be predicted by variables $x_1, x_2, \ldots, x_k$. In order to distinguish values, $\hat{y}$ expresses predicted values and y expresses observed values. Here, $x_1, x_2, \ldots, x_k$ are called independent variables and y is called dependent variable.

3.2.2. Determining fuzzy set and dependent variable

The rule of determining regression parameters is to make the sum of the square of dispersion between $\hat{y}$ and y reach the minimum. The method to be used is least squares method and the expression is given as:

$$\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \min \qquad (8)$$

According to the expression (8), regression parameters can be gained and then the regression model can be preliminarily established. For observed values $x_{i1}, x_{i2}, \ldots, x_{ik}$ of any group of independent variables, there exist $\hat{y}$ and y accordingly. So a fuzzy set is defined and the universe of discussion is all the data points. The fuzzy concept is the deviation degree between all the data points and fitness equations. The farther a data point is apart from the fitness equation, the smaller the membership degree is. On the contrary, the closer a data point is near the fitness equation, the bigger the membership degree is. Because the value of random variables usually conforms to normal distribution, the membership function can be given as the following form:

$$u_i = e^{-(\frac{y_i - \hat{y}_i}{\sigma})^2}, i = 1, 2, \ldots, n, \sigma \in R \qquad (9)$$

3.2.3. Determining regression coefficients with fuzzy least squares method

If we add fuzzy domain to least squares method, the effect of discrete data points on the fitness result will be reduced and the effect of concentrated data points on the fitness result will be enhanced. Such is called fuzzy least squares method. The fuzzy least squares method is represented by the following expression:

$$\sum_{i=1}^{n} u_i (y_i - \hat{y}_i)^2 = \min \qquad (10)$$

When $u_i = 1$, the expression (10) will become the least squares method. On the basis of the expression (10), a group of equations to obtain the fuzzy parameters are provided as:

$$\begin{cases} s_{11}b_1 + s_{12}b_2 + \cdots + s_{1k}b_k = s_{1y} \\ s_{21}b_1 + s_{22}b_2 + \cdots + s_{2k}b_k = s_{2y} \\ \cdots \\ s_{k1}b_1 + s_{k2}b_2 + \cdots + s_{kk}b_k = s_{ky} \end{cases} \qquad (11)$$

Here, $s_{ij} = \sum u \sum u x_i x_j - \sum u x_i \sum u x_j$, and $s_{iy} = \sum u \sum u x_i y - \sum u x_i \sum u y$, $i, j = 1, 2, \cdots, k$. According to this group of equations, first we can obtain the values of variables $b_1, b_2, \cdots, b_k$. And finally b_0 is gained by:

$$b_0 = \frac{\sum uy}{\sum u} - b_1 \frac{\sum ux_1}{\sum u} - b_2 \frac{\sum ux_2}{\sum u} - \cdots - b_k \frac{\sum ux_k}{\sum u} \qquad (12)$$

Thus, the first fuzzy regression is finished, i.e. the first iteration. Then return to the expression (9), determine the membership functions of fuzzy sets with the new coefficients and gain the coefficients of expression (7) finally. Such is the second iteration. Little by little, compared with the last iteration, the quantity grades of regression parameters gained after several iterations are very close, i.e. the difference between the quantity grades is tiny. At that time, the final regression model can be established.

3.2.4. Coefficient of determination

Coefficient of determination is used to assess the quality of the estimation models and expressed by R^2. The coefficient R^2 is calculated by:

$$R^2 = \frac{\sum_{i=1}^{n}(y - \overline{y})^2}{\sum_{i=1}^{n}(\hat{y} - \overline{y})^2} \qquad (13)$$

Here, $\overline{y}$ expresses the mean value of random variables. Obviously, the coefficient R^2 describes the percentage of variability and the value is between 0 and 1; when an R^2 is close to 1, it indicates that this model can explain variability in the response to the predictive variable, i.e. there is a strong relationship between the independent and dependent variables.

4. The application of the fuzzy decision-making model to COSMIC-FFP

The COSMIC-FFP size estimation method accounts for all the functionality within a real-time system. The

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efficacy of the COSMIC-FFP can be completely exploited only when it is applied in effort estimation models early in the software lifecycle. But during the late lifecycle, only linear and non-linear regression models for effort estimation with COSMIC-FFP have been explored [6]. In this paper, the data sets in [6] are used to develop fuzzy decision-making regression models. In the linear and non-linear regression models, there is also a lot of uncertainty in the modeling because the other factors affecting the software effort have not been identified. Thus the applicability of fuzzy decision-making models based on fuzzy regression is justified.

4.1. Linear and non-linear regression models

The simple linear regression model with one independent variable size is presented in [6]. Then multiple regression models were produced and these models took into account additional variables that could have an influence on the productivity of software projects.

With the full data set of 15 maintenance projects of organization A [6], the simple linear regression model with one independent variable functional size was constructed in Figure 1 [6]. The model yields $Effort = 91.161 + 0.768 \times Cfsu$, where Cfsu is a standard unit of measurement, i.e. one COSMIC-FFP functional size unit. This model has a coefficient of determination (R2) of 0.38, which indicates that only 38% of the total variability of the dependent variable effort is explained by the variability of the independent variable functional size.

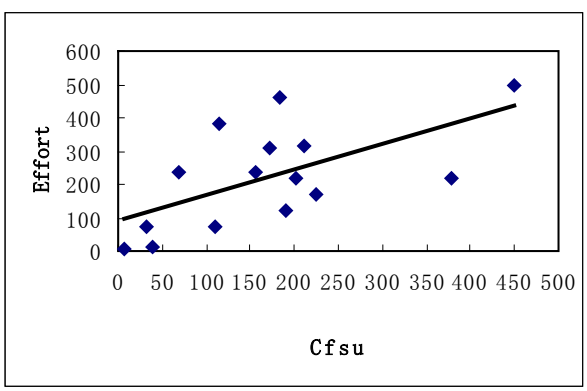

Figure 1. Linear regression model on functional size of organization A

This linear regression model provides an initial indication that either the relationship might not be linear or that another variable might have a significant influence on the relationship between size and work effort [9][10].

Besides the linear regression models, various other non-linear regression models were investigated, as illustrated in Table 1. Like the above, this table is based on the same 15 observations. From the Table 1 [6], we can come to a conclusion that the relationship between the increase in effort due to an increase in size is not strong.

Table 1. Non-linear regression models [6]

Equation	A	B	R	R^2
$Y=A\times X^B$	0.94	1.046	0.846	0.716
$Y=A\times e^{(B\times X)}$	45.95	0.006	0.662	0.387
$Y=A+B\times \ln(X)$	-241.20	96.792	0.664	0.442
$Y=A+B/X$	265.16	-2570.562	0.524	0.274
$Y=1/(A+B\times X)$	0.057	-0.0002	0.477	0.228

4.2. Applying the fuzzy decision-making model to the data set

The COSMIC-FFP size-effort relation is modeled with the fuzzy decision-making model based on fuzzy regression. The principle of fuzzy regression is on the basis of fuzzy expectation and fuzzy variance.

The data set collected at organization A, which develops Web-based applications, is presented in Table 2 [6]. Functional size varies from 8 Cfsu to 450 Cfsu, with an average of 169 Cfsu, while effort varies from 5 hours to 495 hours, with an average of 221 hours (or 1.5 person/hour).

Table 2. The data set of organization A

Projects	1	2	3	4	5
Cfsu	111	397	190	157	202
Effort(h)	70	219	119	238	216
Projects	6	7	8	9	10
Cfsu	224	183	8	115	32
Effort(h)	167	460	5	383	70.5
Projects	11	12	13	14	15
Cfsu	450	39	172	210	69
Effort(h)	495.25	15	312	314.5	235

Here, we consider Cfsu as a dependent variable and effort as an independent variable. From the Table 2, we can approximately find the data of several projects are exceptional. It indicates that there exists morbid data in the sample points. Now, we establish a regression model for the sample universe. In order to make the value of membership functions spread between 0 and 1 properly, we define the membership function as the following expression:

$$u_i = e^{-(\frac{\hat{y}_i - y_i}{10})^2}, i = 1, 2, \ldots, 15 \qquad (14)$$

Of course, this expression is proved to be effective and completely conforms to the definition of membership functions.

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Based on the observed values (Table 2), we should determine the regression parameters. Because we only take the functional size into account, we only need calculate b_0 and b_1 . So the regression model is represented as:

$$\hat{Effort} = b_0 + b_1 \times Cfsu \qquad (15)$$

We can gain the value of b_1 based on equation (11), and then calculate the value of b_0 according to equation (12).

Firstly, we can obtain the result of the zeroth regression by least squares method. In fact, the typical linear regression is just the zeroth regression. Then we can use the expressions (11) and (14) repeatedly to gain other regression models. Of course, each regression model is based on the last regression model. Because we must calculate the value of each $\hat{y}_i$ according to the last regression equation first, we may get the u_i based on expression (14). In this way, we calculate 18 regression equations. These iterative results are as follows:

The result of the zeroth regression is:

$$\hat{Effort} = 91.161210 + 0.768135 \times Cfsu \ .$$

The result of the first regression is:

$$\hat{Effort} = 46.998469 + 0.977563 \times Cfsu \ .$$

The result of the second regression is:

$$\hat{Effort} = 22.790321 + 1.064611 \times Cfsu \ .$$

…

The result of the fifteenth regression is:

$$\hat{Effort} = 1.073694 + 1.081019 \times Cfsu \ .$$

The result of the sixteenth regression is:

$$\hat{Effort} = 1.065936 + 1.081011 \times Cfsu \ .$$

The result of the seventeenth regression is:

$$\hat{Effort} = 1.061344 + 1.081006 \times Cfsu \ .$$

From the above regression lines, it is obvious that these regression parameters are very close after the fifteenth regression. The last two regression equations are very similar.

5. Results

From the above results, we can find that the regression model tends to be steady little by little. When the regression model is almost steady, the final regression model can be determined. This model is the closest to the real situation. In practice, the regression frequency can be controlled according to actual needs. In Figure 2, the first, the second and the last regression lines are drawn in the same reference frame. The regression lines tend to descend because the values of several sample points in the sample universe are higher than the real value.

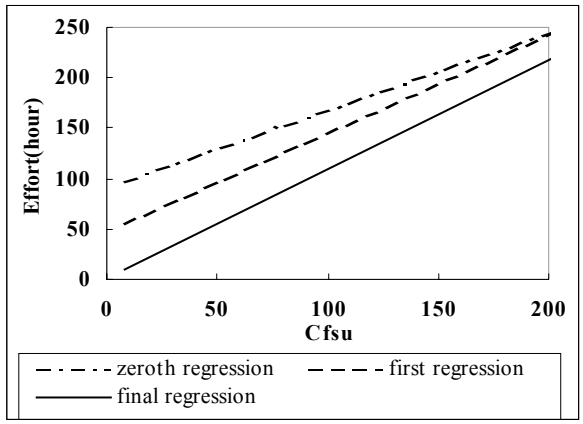

Figure 2. Regression lines distribution

In Figure 3, the membership degree of several regressions is drawn. We will find that the membership degree of the 2nd, 7th, 9th data points is very small and close to 0. Thus the 3 points lie away from the regression lines so that these data points are unimportant for making decision. Maybe there are other factors that affect the effort of projects. After several regressions, a best group of membership degrees can be gained.

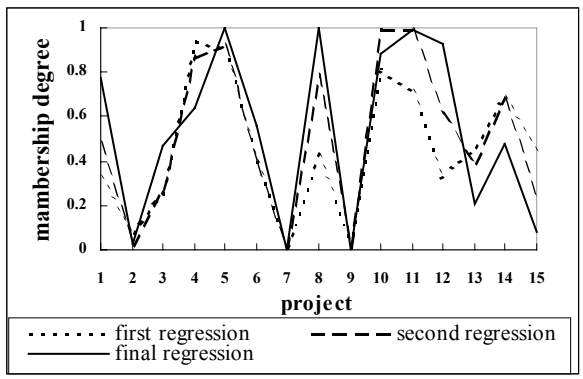

Figure 3. Membership degree distribution

By calculation, the coefficient R2 is 0.813, which indicates that 81.3% of the total variability of the dependent variable effort is explained by the variability of the independent variable functional size. This result is better than the linear, non-linear and other fuzzy regression models. So this model is useful for us.

6. Conclusions

The goal of this paper is to investigate the development of estimation model for maintenance

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projects in order to model the relationship between the size of software of a maintenance project and the maintenance work effort. We use the functional size of software enhancements as inputs to output the effort we should provide. As a whole, the effort required to deliver a functional enhancement in the maintenance phase increases with the size of the enhancement software. In this paper, a fuzzy decision-making model has been established. And a proper membership function has been defined according to the normal distribution characteristic of random variables.

The results prove that this model is better than linear regression and non-linear models. Compared with these traditional regression models, this model can make the disturbance of morbid data and bad information in regression analysis to the result decrease little by little, until we are satisfied. And also this model can determine regression frequency according to actual needs.

As we all know, Software size is recognized as a key factor in the construction of models to estimate project effort. Of course, software size can be measured from either a technical perspective, with LOC (lines of code) for example, or from a functional perspective. The COSMIC-FFP method can measure the functional size of almost all the software types. In the early stage of software development, we can measure the functional size from the specifications and then estimate the effort to be spent based on the fuzzy decision-making model. But in practice, some other factors may influence the effort, such as project difficulty, program language and LOC etc. So in the late stage of software development (e.g. maintenance stage), we can take these factors into account according to the expression (7) and build up a new model. In this way, the result will be more accurate and more useful and the coefficient R2 will be increased.

In the practical applications, if math fuzzy is applied into the software metrics by making fuzzy sets participate in decision-making, it can raise accuracy and reliability of the decision result.

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